

The smooth-operator subalgebra problem was answered negatively in 2015

Literature-status note for arXiv:1304.7189

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Status: literature_already_answered. Open Problem 4.13 of Ciaś (2013) asks whether every closed commutative $*$ -subalgebra of $L(s', s)$ is isomorphic to a closed $*$ -subalgebra of the pointwise algebra s . Ciaś (2015), Theorem 6.10, explicitly states that it solves this problem in the negative and constructs a counterexample.

Original problem

Let s be the Fréchet space of rapidly decreasing sequences and let $L(s', s)$ be the Fréchet $*$ -algebra of continuous operators from its strong dual s' into s . In *On the algebra of smooth operators*, Ciaś asks the following on page 16 of the local source compilation [1, Open Problem 4.13]:

Is every closed commutative $*$ -subalgebra of $L(s', s)$ isomorphic to some closed $*$ -subalgebra of s with pointwise multiplication?

The source gives an equivalent Köthe-algebra formulation using an s -complete orthonormal system (e_n) and $\lambda^\infty(|e_n|_q)$.

Explicit later answer

The separate later paper *Commutative subalgebras of the algebra of smooth operators* says immediately before its decisive theorem that the next result “solves in negative” Open Problem 4.13 of the 2013 paper [2, p. 14]. Its Theorem 6.10 states:

There is a closed commutative $*$ -subalgebra of $L(s', s)$ which is not isomorphic to any closed $*$ -subalgebra of s .

This is a direct counterexample to the universal assertion in the original problem, not merely an implication identified after the fact.

The later paper also gives the structural boundary. Theorem 6.2 proves that an infinite-dimensional closed commutative $*$ -subalgebra $E \subset L(s', s)$ is isomorphic to a closed $*$ -subalgebra of s exactly when E has Vogt’s property (Ω) (equivalently, when it is isomorphic as a Fréchet space to a complemented subspace of s). In the notation of its canonical basis, this is further equivalent to an explicit polynomial domination condition on the associated weights. Theorem 6.10 then constructs a system whose weights violate that condition.

Scope and provenance

The 2013 yes/no problem is fully answered: the answer is *no*. The 2015 paper supplies both a characterization of the positive cases and an explicit negative example. The supporting author is the author of the source paper, cites the source as reference [3], and explicitly identifies Open Problem 4.13 as the problem being solved. This therefore belongs in the literature-already-answered category and is not a new result of this run.

The identification was checked against the official ingested TeX of both papers, their locally compiled PDFs, the current arXiv records, and the run's registry, solution, attempt, and proof-gap indexes. The packet contains both local PDFs.

References

- [1] Tomasz Ciaś, *On the algebra of smooth operators*, Studia Math. **218** (2013), no. 2, 145–166; arXiv:1304.7189v1.
- [2] Tomasz Ciaś, *Commutative subalgebras of the algebra of smooth operators*, arXiv:1503.07095v1, 2015.